

✓ Project: Smol Learning Agent — Auto-Generated YouTube Learning Plan

Overview

This project implements a fully automated Learning Agent using SmolAgents.

It takes any topic (e.g., SQL, Generative AI, Data Science), searches the web for the best YouTube playlist using DuckDuckGo, and then creates:

- The best free playlist link
- A 12-hour learning assumption
- A completion date (1 hour/day)
- A 12-day structured study plan

All steps are automated inside a single notebook.

✓ Step 1 — Install required packages

We install SmolAgents and DuckDuckGo Search library.

```
1 !pip install smolagents -U
```

```
Requirement already satisfied: smolagents in /usr/local/lib/python3.12/dist-packages (1.23.0)
Requirement already satisfied: huggingface-hub<1.0.0,>=0.31.2 in /usr/local/lib/python3.12/dist-packages (from smolagents) (0.36.0)
Requirement already satisfied: requests>=2.32.3 in /usr/local/lib/python3.12/dist-packages (from smolagents) (2.32.4)
Requirement already satisfied: rich>=13.9.4 in /usr/local/lib/python3.12/dist-packages (from smolagents) (13.9.4)
Requirement already satisfied: jinja2>=3.1.4 in /usr/local/lib/python3.12/dist-packages (from smolagents) (3.1.6)
Requirement already satisfied: pillow>=10.0.1 in /usr/local/lib/python3.12/dist-packages (from smolagents) (11.3.0)
Requirement already satisfied: python-dotenv in /usr/local/lib/python3.12/dist-packages (from smolagents) (1.2.1)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<1.0.0,>=0.31.2->smolagents) (3.20.0)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<1.0.0,>=0.31.2->smolagents)
Requirement already satisfied: packaging>=20.9 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<1.0.0,>=0.31.2->smolagents) (
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<1.0.0,>=0.31.2->smolagents) (6.0.
Requirement already satisfied: tqdm>=4.42.1 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<1.0.0,>=0.31.2->smolagents) (4.6
Requirement already satisfied: typing-extensions>=3.7.4.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<1.0.0,>=0.31.2->sm
Requirement already satisfied: hf-xet<2.0.0,>=1.1.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<1.0.0,>=0.31.2->smolagen
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2>=3.1.4->smolagents) (3.0.3)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.3->smolagents) (3.4.4
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.3->smolagents) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.3->smolagents) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.3->smolagents) (2025.10.5)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=13.9.4->smolagents) (4.0.0)
```

Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=13.9.4->smolagents) (2.19.2)
 Requirement already satisfied: mdurl~0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=13.9.4->smolagents) (

✓ Step 2 — Login to HuggingFace

This stores the HF token in the notebook environment.

```
1 from huggingface_hub import notebook_login
2
3 notebook_login()
```

```
1 !pip install ddgs
2
```

```
Requirement already satisfied: ddgs in /usr/local/lib/python3.12/dist-packages (9.9.1)
Requirement already satisfied: click>=8.1.8 in /usr/local/lib/python3.12/dist-packages (from ddgs) (8.3.0)
Requirement already satisfied: primp>=0.15.0 in /usr/local/lib/python3.12/dist-packages (from ddgs) (0.15.0)
Requirement already satisfied: lxml>=4.9.4 in /usr/local/lib/python3.12/dist-packages (from ddgs) (5.4.0)
Requirement already satisfied: httpx>=0.28.1 in /usr/local/lib/python3.12/dist-packages (from httpx[brotli,http2,socks]>=0.28.1->ddgs) (0.28.1)
Requirement already satisfied: fake-useragent>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from ddgs) (2.2.0)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx>=0.28.1->httpx[brotli,http2,socks]>=0.28.1->ddgs) (4.3.0)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx>=0.28.1->httpx[brotli,http2,socks]>=0.28.1->ddgs) (2025.11.12)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx>=0.28.1->httpx[brotli,http2,socks]>=0.28.1->ddgs) (1.0.7)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx>=0.28.1->httpx[brotli,http2,socks]>=0.28.1->ddgs) (3.10.1)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx>=0.28.1->httpx[brotli,http2,socks]>=0.28.1->ddgs) (0.14.0)
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Requirement already satisfied: h2<5,>=3 in /usr/local/lib/python3.12/dist-packages (from httpx[brotli,http2,socks]>=0.28.1->ddgs) (4.3.0)
Requirement already satisfied: socksio==1.* in /usr/local/lib/python3.12/dist-packages (from httpx[brotli,http2,socks]>=0.28.1->ddgs) (1.0.0)
Requirement already satisfied: hyperframe<7,>=6.1 in /usr/local/lib/python3.12/dist-packages (from h2<5,>=3->httpx[brotli,http2,socks]>=0.28.1->ddgs) (6.1.0)
Requirement already satisfied: hpack<5,>=4.1 in /usr/local/lib/python3.12/dist-packages (from h2<5,>=3->httpx[brotli,http2,socks]>=0.28.1->ddgs) (4.1.0)
Requirement already satisfied: sniffio>=1.1 in /usr/local/lib/python3.12/dist-packages (from anyio->httpx>=0.28.1->httpx[brotli,http2,socks]>=0.28.1->ddgs) (1.3.1)
Requirement already satisfied: typing_extensions>=4.5 in /usr/local/lib/python3.12/dist-packages (from anyio->httpx>=0.28.1->httpx[brotli,http2,socks]>=0.28.1->ddgs) (4.12.0)
```

✓ Step 3 — Create a custom tool to find a YouTube playlist

- Uses DuckDuckGo search
- Filters for YouTube links
- Returns the first playlist found

```
1 from smolagents import tool, DuckDuckGoSearchTool
2
```

```
3 @tool
4 def find_youtube_playlist(topic: str) -> str:
5     """
6     Search the web and return a real YouTube playlist for the topic.
7
8     Args:
9         topic: The learning subject (Python, SQL, etc.)
10    """
11    search = DuckDuckGoSearchTool()
12    query = f"{topic} full course playlist YouTube"
13    results = search(query=query)
14
15    # Filter to return the first YouTube playlist link
16    for item in results.split("\n"):
17        if "youtube.com" in item.lower() or "youtu.be" in item.lower():
18            return item
19
20    return "No YouTube playlist found. Try a different topic."
21
```

✓ Step 4 — Build the CodeAgent

- Uses HF Inference API (free model)
- Uses two tools: DuckDuckGo + Playlist Finder
- Allows Python execution inside the agent

```
1 from smolagents import CodeAgent, InferenceClientModel
2
3 agent = CodeAgent(
4     tools=[find_youtube_playlist, DuckDuckGoSearchTool()],
5     model=InferenceClientModel(),
6     additional_authorized_imports=['datetime'],
7     max_steps=5,
8     verbosity_level=2
9 )
10
```

✓ Step 5 — Ask the user what they want to learn

Step 6 — The Agent does everything automatically

1. Searches for playlist

2. Calculates completion date
3. Generates 12-day study plan
4. Prints final answer

```
1 topic = input("What do you want to learn today? ")
2
3 agent.run(f"""
4 I want to learn {topic}.
5
6 Tasks:
7 1. Search the web and give me the best free YouTube playlist.
8 2. Assume playlist total duration = 12 hours.
9 3. If I study 1 hour per day, calculate completion date.
10 4. Give a structured study plan specific to {topic}.
11 """)
12
```


What do you want to learn today? genai

New run

I want to learn genai.

Tasks:

1. Search the web and give me the best free YouTube playlist.
2. Assume playlist total duration = 12 hours.
3. If I study 1 hour per day, calculate completion date.
4. Give a structured study plan specific to genai.

InferenceClientModel - Qwen/Qwen3-Next-80B-A3B-Thinking

Step 1

Output message of the LLM:

Thought: I need to find the best free YouTube playlist for learning GenAI. I'll use the `find\_youtube\_playlist` tool with the topic "generative AI" since that's the standard term for this field.

```
<code>
playlist_url = find_youtube_playlist(topic="generative AI")
print(playlist_url)
```

– Executing parsed code:

```
playlist_url = find_youtube_playlist(topic="generative AI")
print(playlist_url)
```

Execution logs:

[Generative AI Full Course tutorial Playlist -
YouTube](https://www.youtube.com/playlist?list=PLLCcYPd7d0kYrinQKtXFZc86FWd_eEbdu)

Out: None

[Step 1: Duration 29.12 seconds | Input tokens: 2,182 | Output tokens: 4,035]

Step 2

Output message of the LLM:

Thought: Now that I have the YouTube playlist URL from the previous step, I need to calculate the completion date assuming a 12-hour total duration with 1 hour of study per day. Then, I'll create a structured study plan covering key GenAI topics over 12 days. I'll use Python's datetime module to compute the date and format the study plan as a list of daily topics.

```
<code>
import datetime

today = datetime.datetime.now()
completion_date = today + datetime.timedelta(days=12)
formatted_date = completion_date.strftime("%Y-%m-%d")
```

```
study_plan = [
```